

MENTAL WORKLOAD DETECTION USING EEG SPECTRAL FEATURES

22AIE213 : Machine Learning

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PROBLEM STATEMENT

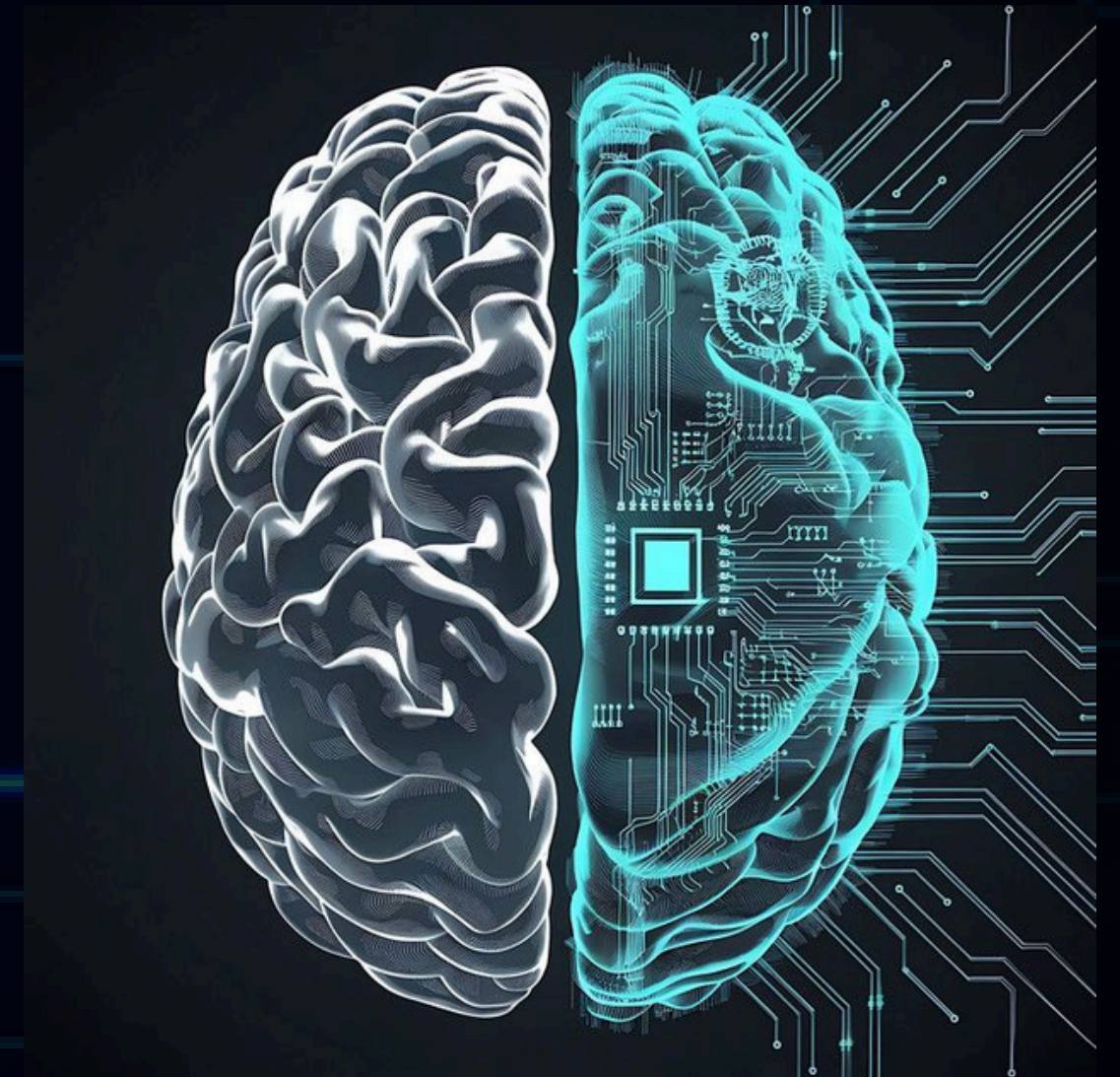
- Traditional workload assessment methods are subjective and lack real-time precision
- Raw EEG signals are complex, making it difficult to reliably detect cognitive workload
- There is a need for an automated EEG-based system to classify workload-present vs. workload-absent mental states

INTRODUCTION:

- Mental workload represents the cognitive effort required to perform a task.
- Excessive workload can reduce attention, decision-making ability, and task performance.
- EEG (Electroencephalography) is a non-invasive technique used to measure brain electrical activity.
- EEG signals contain important information about cognitive states such as concentration, attention, and mental effort.
- Mental arithmetic tasks produce measurable changes in EEG frequency bands.
- Machine learning techniques can automatically classify workload-related EEG patterns.

OBJECTIVE

- Extract EEG spectral features
- Analyze workload-related EEG activity
- Perform statistical feature selection
- Train multiple ML models
- Compare classifier performance
- Identify most discriminative EEG features



RESEARCH GAP:

Existing Limitations	Proposed Work
Many studies use only basic EEG features.	Extracted multiple EEG spectral features.
Single classifiers struggle with nonlinear EEG patterns	Used statistical feature selection.
Limited statistical feature analysis is performed.	Compared multiple ML models.
EEG signals are noisy and highly variable.	Compared multiple ML models.
Some models show poor generalization capability.	Identified biologically significant workload features

DATASET DESCRIPTION

Dataset Source:

PhysioNet — EEG During Mental

Arithmetic Tasks:

- Open-access EEG database
- Published in 2018
- EEGMAT dataset

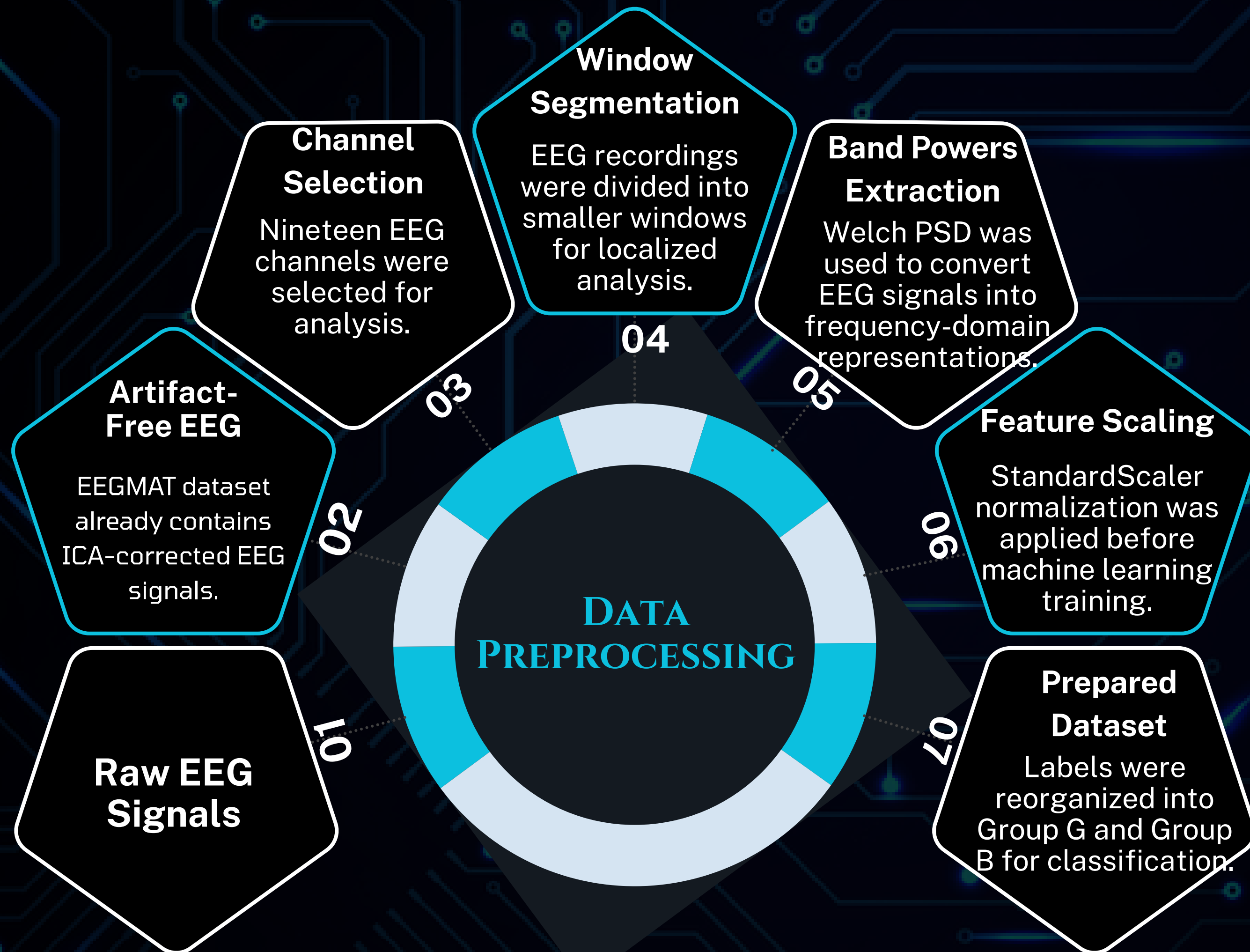
Subjectinfo.csv :

Contains labels of classes 0(B) 1(G)

Citation:

Zyma et al., EEG During Mental
Arithmetic Tasks, PhysioNet, 2018

Parameter	Value
Subjects	35
Channels	23
Recording Type	EEG
File Format	EDF
Sampling Frequency	1000 Hz
Recording Duration	60 Sec
Labels	Bad Count (0) Good Count (1)



WINDOWING TECHNIQUE:

Parameter	Value
Window Size	2 sec
Overlap	50%
Step Size	1 sec

- EEG signals are highly non-stationary and change over time.
- The continuous EEG recordings were divided into smaller windows before feature extraction.
- Window segmentation enables localized analysis of brain activity.
- Each EEG window was processed independently for feature extraction.
- Segmenting the signals increases the number of training samples for machine learning.

FREQUENCY BAND EXTRACTION

Power Spectral Density using Welch Method:

- Welch's method was used to estimate the Power Spectral Density (PSD) of EEG signals.

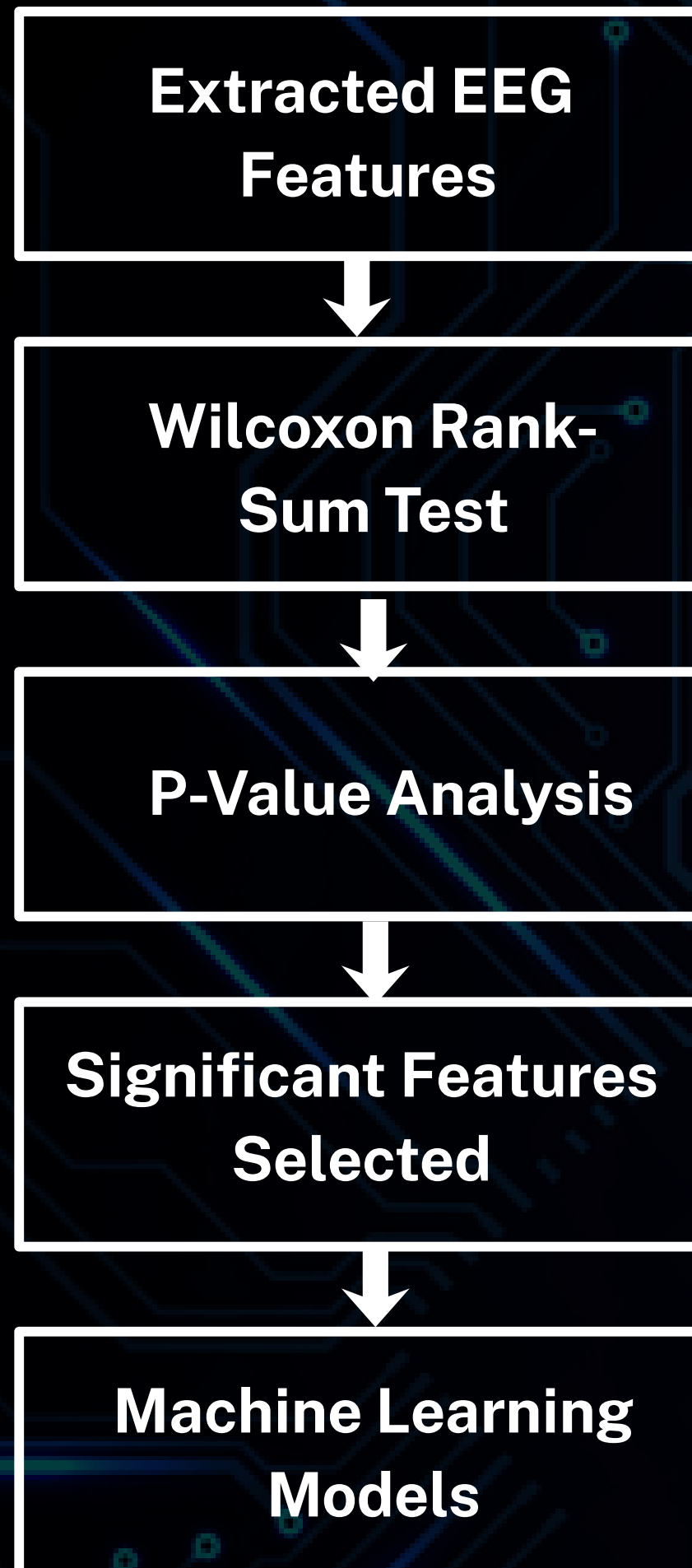
Band	Frequency Range	Brain Activity
Delta	0.5 - 4 Hz	Deep Relaxation
Theta	4 - 8 Hz	Mental Effort
Alpha	8 - 13 Hz	Relaxation
Beta	13 - 30 Hz	Active Thinking
Gamma	30 - 45 Hz	High Cognitive Activity

FEATURE ENGINEERING

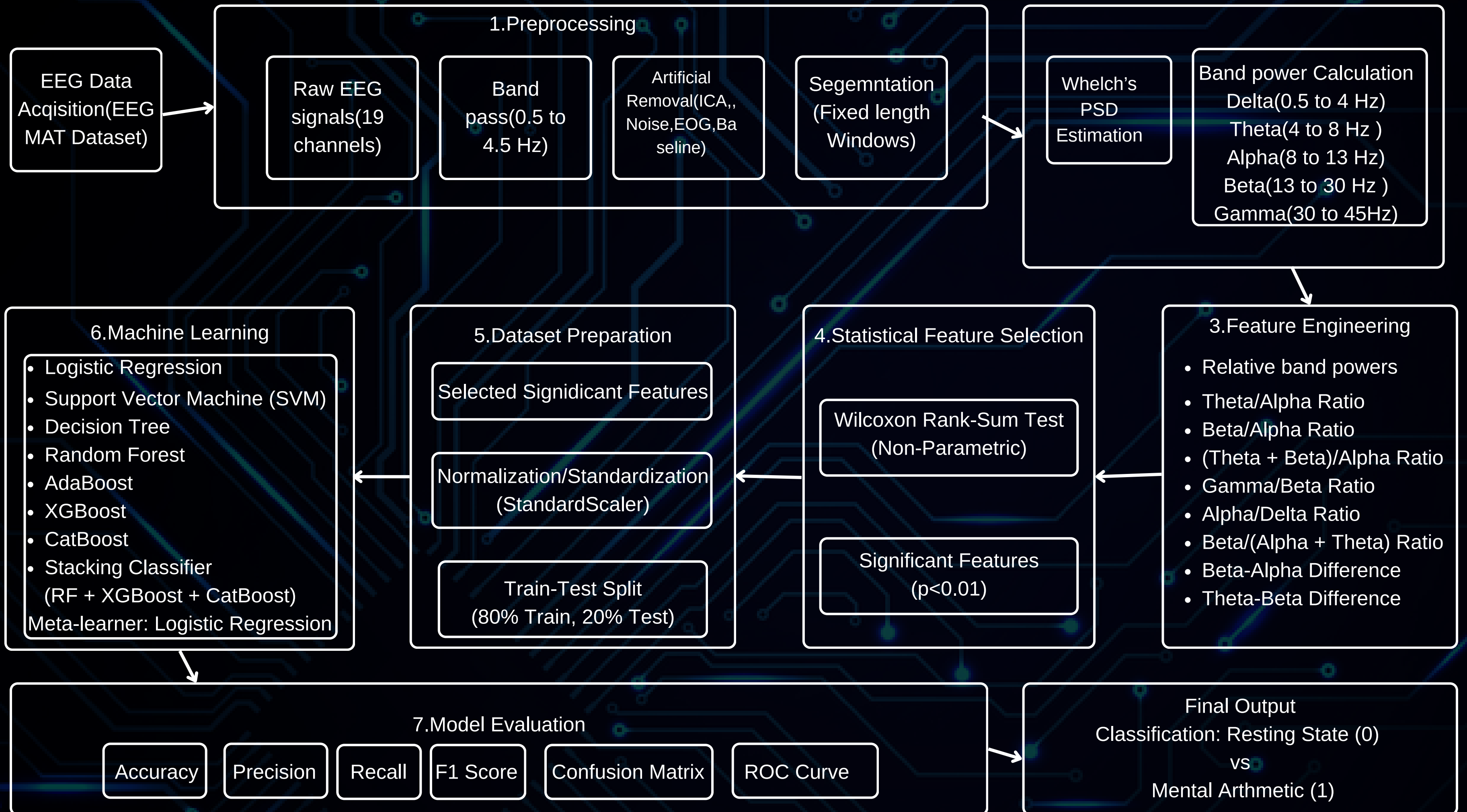
Feature Type	Purpose
Absolute Band Power Features	Capture the strength of neural activity in each EEG frequency band.
Relative Power Features	Measure the contribution of each frequency band relative to total brain activity.
Ratio Features	Capture the balance between concentration, memory processing, and relaxation states.
Difference Features	Measure the dominance of one cognitive state over another during mental tasks.

More than 300 engineered EEG features were extracted.

STATISTICAL FEATURE EXTRACTION:



- Large numbers of EEG features may contain redundant or non-informative information.
- Statistical feature selection helps identify the most discriminative EEG features.
- Wilcoxon Rank-Sum Test was used to compare feature distributions between classes.
- Features with $p\text{-value} < 0.01$ were selected as statistically significant.
- A total of 252 significant EEG features were selected for classification.



MACHINE LEARNING MODELS USED:

Model	Why It Was Used	Why It Worked / Limitation
Logistic Regression	Used as a baseline linear classifier	EEG signals are nonlinear, so performance was limited
SVM	Effective for high-dimensional EEG data	Achieved high recall and strong workload detection
Decision Tree	Simple and interpretable model	Sensitive to noise and prone to overfitting
Random Forest	Uses multiple trees to improve robustness	Reduced overfitting and improved precision

MACHINE LEARNING MODELS USED:

Model	Why It Was Used	Why It Worked / Limitation
AdaBoost	Sequentially improves weak learners	Moderately effective on complex EEG patterns
XGBoost	Learns complex nonlinear feature relationships	Captured workload-related EEG patterns effectively
CatBoost	Improves generalization and reduces overfitting	Performed well on complex spectral features
Stacking Classifier	Combines multiple strong classifiers	Achieved best overall classification performance

EVALUATION METRICS:

Metric	What it measures:	Why It Is Important	Formula
Accuracy	Overall correctness of the model	Measures total correctly classified EEG samples	$(TP + TN) / \text{Total}$
Precision	Reliability of workload predictions	Reduces false workload detections	$TP / (TP + FP)$
Recall	Ability to detect actual workload states	Important for not missing workload samples	$TP / (TP + FN)$
F1-Score	Balance between precision and recall	Gives balanced classification performance	$2 \times (\text{Precision} \times \text{Recall}) / (\text{Precision} + \text{Recall})$
ROC-AUC	Ability to separate classes	Measures overall discriminative capability	Area Under the curve

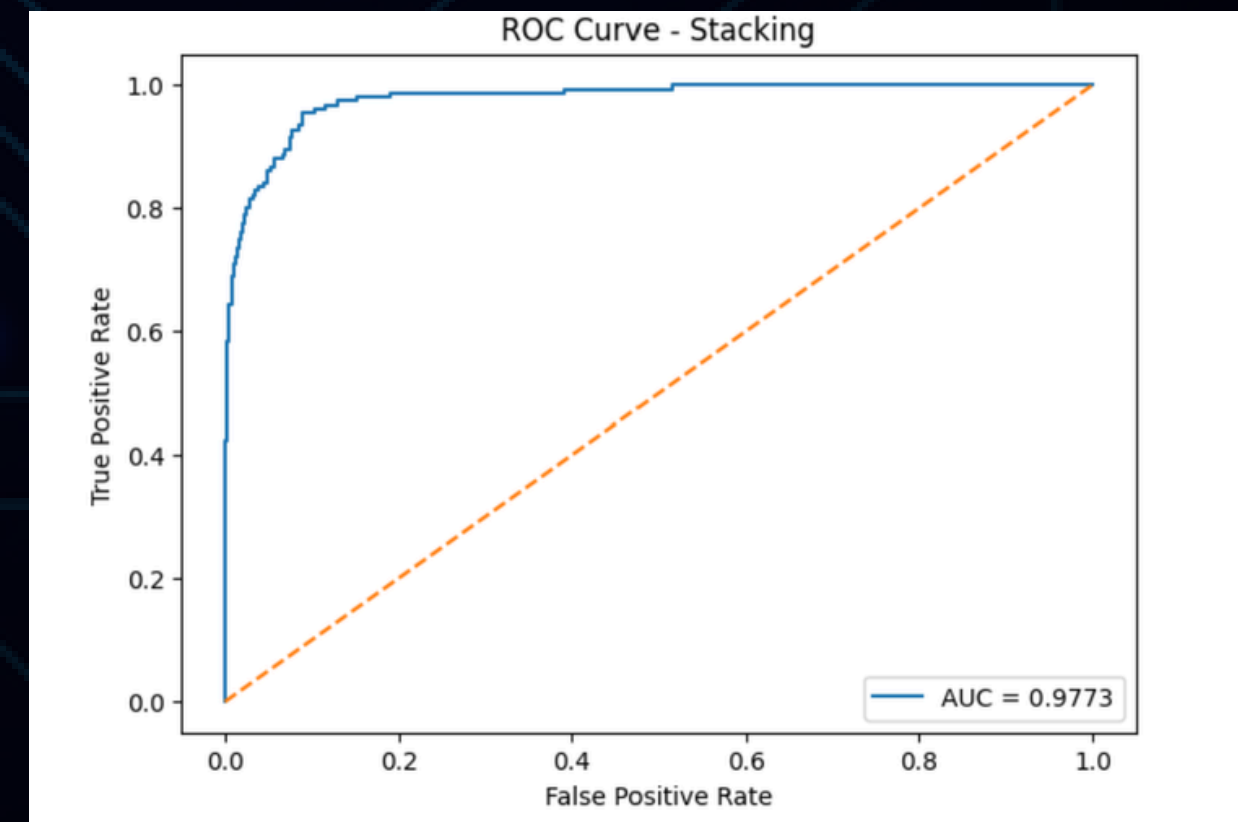
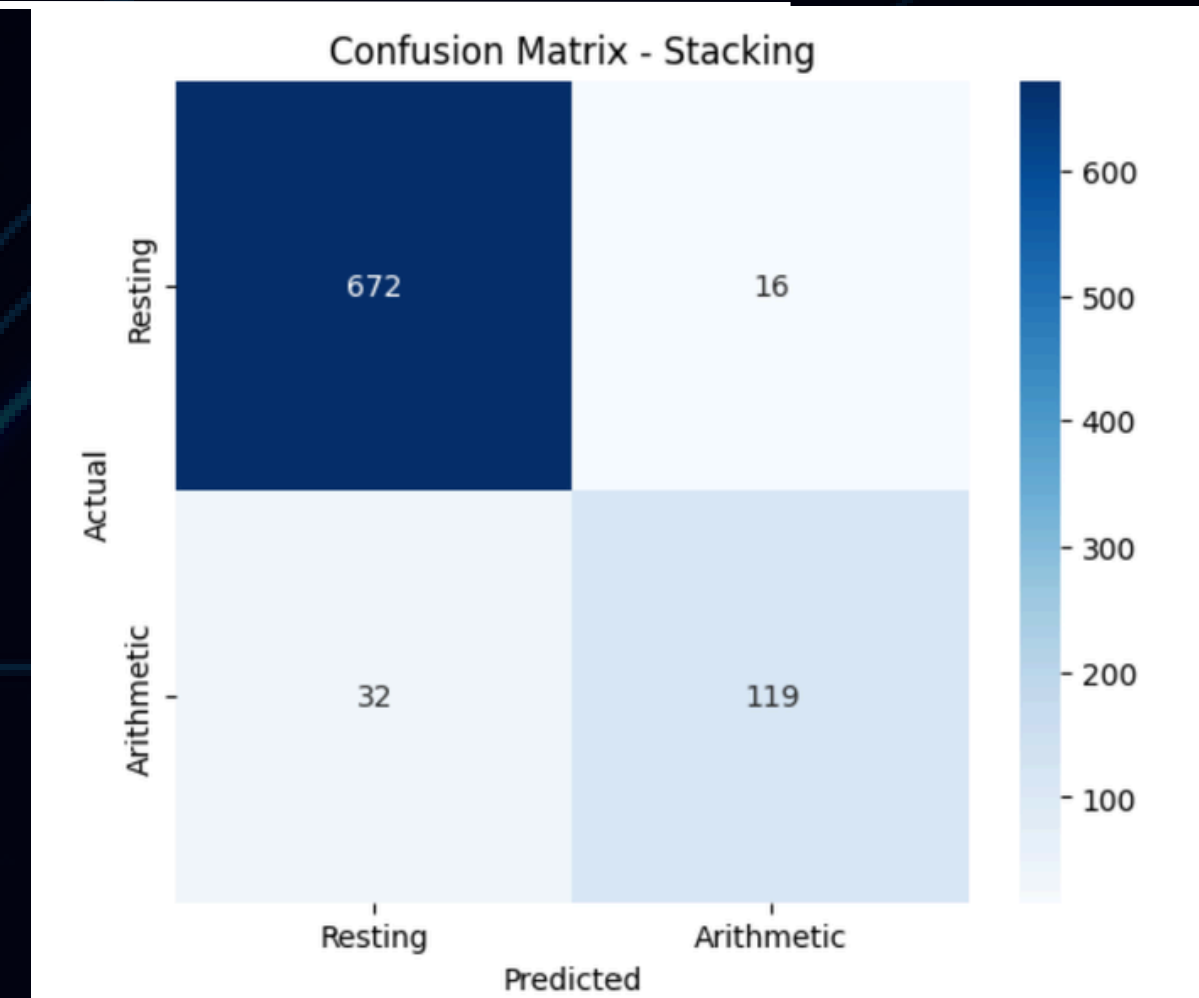
ML MODEL RESULTS COMPARISON:

Model	Accuracy	Precision	Recall	F1 Score	ROC
Adaboost	85.46	0.7231	0.3113	0.4352	0.8693
Random Forest	89.27	0.9692	0.4172	0.5833	0.9667
Ada Boost	90.58	0.9286	0.5166	0.6638	0.9682
XG Boost	92.97	0.9423	0.6490	0.7686	0.9751
Stacking	94.28	0.8815	0.7881	0.8322	0.9773

BEST MODEL - STACKING CLASSIFIER:

- Base Learners:
 - Random Forest
 - XGBoost
 - CatBoost
- Meta Learner:
 - Logistic Regression
- Combines predictions from multiple strong classifiers
- Improves robustness and generalization
- Reduces individual model weaknesses

Accuracy	Precision	Recall	F1 Score	ROC
0.9428	0.8815	0.7881	0.8322	0.9773



FUTURE WORKS:

- Development of real-time EEG workload monitoring systems.
- Implementation of deep learning models such as CNNs and LSTMs.
- Cross-subject workload classification for improved generalization.
- Integration with brain-computer interface (BCI) applications.
- Use of larger EEG datasets for improved robustness.
- Deployment in healthcare, driver monitoring, and adaptive learning systems.

SOTA:

Study	Task	Classifier	Accuarcy
Ref[93]	N-back	KNN, RF, DT, SVM	84.6%
Ref[32]	Raven test	LR, RF, XGB	71.3%
Ref[91]	Arthematic test	SVM, KNN, DT, RF	91.0%
Ref[12]	Real ATM	LR, LDA, SVM, KNN	84.0%
Ref[95]	Simulated flight	KNN, LDA, DT, SVM	90.1%
Proposed work	Mental arthematic EEG	Stacking Ensemble	94.28%

Ref[93]-

B. S. Cheema, S. Samima, M. Sarma, and D. Samanta, "Mental Workload Estimation from EEG Signals Using Machine Learning Algorithms," *Engineering Psychology and Cognitive Ergonomics*. pp. 265-284, 2018.

Ref[32]-

N. Friedman, T. Fekete, K. Gal, and O. Shriki, "EEG-Based Prediction of Cognitive Load in Intelligence Tests," *Frontiers in Human Neuroscience*, vol. 13, no. 191, 2019

Ref[91]-

M. Plechawska-Wójcik, M. Tokovarov, M. Kaczorowska, and D. Zapaa, "A Three-Class Classification of Cognitive Workload Based on EEG Spectral Data," *Applied sciences*, vol. 9, no. 24, pp. 5340, 2019.

Ref[12]-

N. Sciaraffa, P. Aricò, G. Borghini, G. D. Flumeri, A. D. Florio, and F. Babiloni, "On the Use of Machine Learning for EEG-Based Workload Assessment: Algorithms Comparison in a Realistic Task," *Human Mental Workload: Models and Applications*, pp. 170-185, 2019.

Ref[95]-

J. A. Blanco, M. K. Johnson, K. J. Jaquess, H. Oh, L. Lo, R. J. Gentili, and B. D. Hatfield, "Quantifying Cognitive Workload in Simulated Flight Using Passive, Dry EEG Measurements," *IEEE Transactions on Cognitive and Developmental Systems*, vol. 10, no. 2, pp. 373-383, 2018

CONCLUSION:

- EEG spectral features were successfully extracted using Welch PSD analysis.
- Statistical feature selection identified significant workload-related EEG features.
- Multiple machine learning models were evaluated for mental workload classification.
- Ensemble learning models outperformed traditional classifiers.
- The Stacking Classifier achieved the best performance with:
 - 94.28% Accuracy
 - 97.73% ROC-AUC
- Beta-band related EEG features showed strong association with mental workload.



THANK YOU !!